

Comprehensive UAT Test Cases & Requirement Traceability Matrix

E-Learning Platform & Binary MLM System

Document Version: 1.0.0 **Project Root Directory:** `/Users/apple/Desktop/shamshad/e learning`
Scope: Web Application (`/client`), Mobile Android App (`/app`), and REST API Server (`/server`)

1. UAT Strategy & Execution Framework

1.1 Objective

This User Acceptance Testing (UAT) suite ensures that all functional, visual, structural, and financial business requirements defined in the E-Learning & Binary MLM Specification are fully verified before production deployment.

1.2 Testing Environment Setup

- **Web Client:** Vite React + TypeScript (`http://localhost:3000`)
- **API Backend:** Node.js Express (`http://localhost:5000`)
- **Database:** MongoDB (`mongodb://localhost:27017/elearning_db`)
- **Mobile Android App:** React Native Android Emulator / Physical Device (`RZCY900D0JT`)

2. End-to-End UAT Test Case Matrix

Suite 1: Authentication, Localization & User Management

TC-AUTH-01: Student Registration with Referral Sponsor Code

- **Feature Area:** User Management & Binary Placement
- **Requirement Ref:** Section 3.1 & Section 4.1
- **Pre-conditions:** Backend server running; valid referral code available (e.g. `EDU-99201`).
- **Test Steps:**

1. Open Signup page (`/register?ref=EDU-99201`). 2. Enter Full Name ("Amit Verma"), Mobile Number ("9876500001"), State ("Karnataka"), and Password ("Password@123"). 3. Click **"Register Account"**.

- **Expected Outcome:** User registered successfully; JWT token generated; new user automatically placed in sponsor's binary MLM tree under selected leg preference (`AUTO` / `LEFT` / `RIGHT`).
- **Pass/Fail Criteria:** HTTP 201 response; user record created in MongoDB; `MLMNode` created with valid `parent` and `sponsor` fields.

TC-AUTH-02: User Login via Mobile / User ID / Name

- **Feature Area:** Authentication
- **Requirement Ref:** Section 3.1
- **Pre-conditions:** Account created in TC-AUTH-01.
- **Test Steps:**

1. Navigate to `/login` . 2. Input Mobile Number (`9876500001`) or User ID (`EDU-99201`) and Password. 3. Submit login form.

- **Expected Outcome:** Successful authentication; token saved to `localStorage` / SecureStore; redirected to main dashboard.
- **Pass/Fail Criteria:** User profile loaded with correct state localization and wallet balance.

TC-AUTH-03: Regional State Localization Switching

- **Feature Area:** Localization
- **Requirement Ref:** Section 3.1 & Section 3.2
- **Pre-conditions:** Logged in as student.
- **Test Steps:**

1. Click **"Change State Preference"** in header banner. 2. Select state option (e.g., "Karnataka (KA)" or "Delhi NCR (DL)"). 3. Confirm selection.

- **Expected Outcome:** Dashboard dynamically updates hero banners, student count (e.g., "1.2 Lakh+ enrolled in Karnataka"), state-specific board courses (SSLC / KAS prep).
- **Pass/Fail Criteria:** State badge updates instantly; course listings reflect selected state code.

TC-AUTH-04: 1-Year Course Subscription Validity Enforcement

- **Feature Area:** Subscription Management
- **Requirement Ref:** Section 3.1
- **Pre-conditions:** Student enrolling in course batch.
- **Test Steps:**

1. Purchase annual pass for "CBSE Class 10 Master Course". 2. Inspect purchase record in MongoDB database (`purchases` collection).

- **Expected Outcome:** `validUntil` timestamp set to exactly **365 days** from purchase date (`Date.now() + 365 24 60 60 1000`).

- **Pass/Fail Criteria:** Course status active for 1 full year.

Suite 2: KYC Identity Verification & Upload Module

TC-KYC-01: Aadhaar Card Number Format Validation

- **Feature Area:** KYC Module
- **Requirement Ref:** Section 3.1
- **Pre-conditions:** Student logged in; navigating to KYC tab.
- **Test Steps:**

1. Select Document Type: **Aadhaar Card**. 2. Enter invalid Aadhaar number (e.g. `1234-ABCD`). 3. Click **"Submit KYC"**.

- **Expected Outcome:** Validation error displayed: *"Invalid Aadhaar Card Number! Must be exactly 12 numeric digits."* Form submission blocked.
- **Pass/Fail Criteria:** Regex validation blocks non-12-digit numeric input.

TC-KYC-02: PAN Card Format Validation

- **Feature Area:** KYC Module
- **Requirement Ref:** Section 3.1
- **Pre-conditions:** Student logged in.
- **Test Steps:**

1. Select Document Type: **PAN Card**. 2. Enter invalid PAN number (e.g. `12345ABCDE`). 3. Click **"Submit KYC"**.

- **Expected Outcome:** Validation error displayed: *"Invalid PAN Card Number! Format must be 5 Letters + 4 Digits + 1 Letter."*
- **Pass/Fail Criteria:** Regex `^[A-Z]{5}[0-9]{4}[A-Z]{1}$` strictly enforced.

TC-KYC-03: Mobile Camera/Gallery Document Image Upload

- **Feature Area:** KYC Module & Native Mobile App
- **Requirement Ref:** Section 3.1
- **Pre-conditions:** Mobile app running on device/emulator.
- **Test Steps:**

1. Enter valid Aadhaar (`9900 1234 5678`). 2. Click " **Pick Document Image**". 3. Select photo from device image gallery. 4. Submit KYC request.

- **Expected Outcome:** Image uploaded to Cloudinary/server storage; status updated to `PENDING` approval.
- **Pass/Fail Criteria:** KYC record saved in backend with `documentImage` URL and `PENDING` state.

Suite 3: 6 Core Educational Verticals & Content Drilldown

TC-EDU-01: Vertical 1 — School Education (K-12) Navigation

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module I)
- **Test Steps:** Click "**School Education (K-12)**" card → Verify State Board, CBSE, ICSE, PUC (+1/+2 Arts/Commerce/Science) categories.
- **Expected Outcome:** Displays all 24 K-12 courses with grade levels and subjects.

TC-EDU-02: Vertical 2 — School Entrance & Competitive Exams

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module II)
- **Test Steps:** Select "**School Entrance & Competitive**" → Verify Navodaya Entrance, NTS, G-MAT, NET, NEET, JEE batches.
- **Expected Outcome:** Displays NEET & JEE ultimate crash course packages with 5000+ MCQs.

TC-EDU-03: Vertical 3 — Higher Education (UG & PG)

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module III)
- **Test Steps:** Click "**Higher Education (UG & PG)**" → Verify M.A, M.Com, M.Sc, MBA, Custom degree modules.
- **Expected Outcome:** Displays degree courses with subject credits and chapter notes.

TC-EDU-04: Vertical 4 — State Government Jobs Preparation

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module IV)
- **Test Steps:** Select "**State Government Jobs Prep**" → Verify SDA, FDA, GPT (Primary Teacher), KAS Officer prep.
- **Expected Outcome:** Displays KAS Officer General Studies & solved question archives.

TC-EDU-05: Vertical 5 — Central Government Jobs Preparation

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module V)
- **Test Steps:** Select "**Central Government Jobs Prep**" → Verify Banking (IBPS/SBI), Railway RRB, IAS / Civil Services.
- **Expected Outcome:** Displays IBPS PO, RRB NTPC, and UPSC GS Paper I & II prep batches.

TC-EDU-06: Vertical 6 — Teacher Preparation & Certifications

- **Feature Area:** Content Hierarchy
- **Requirement Ref:** Section 2 (Module VI)
- **Test Steps:** Select "**Teacher Prep & Certifications**" → Verify B.Ed, M.Ed, State TET, CTET Paper 1 & 2.

- **Expected Outcome:** Displays teaching methodology, child pedagogy, and solved TET mock tests.

Suite 4: 5 Primary Learning Asset Formats

TC-ASSET-01: Format 1 — Text Reading Lessons

- **Test Steps:** Open Chapter 1 ("Light: Reflection & Refraction") → Click "**Lesson Notes**".
- **Expected Outcome:** Displays formatted reading material and key conceptual rules.

TC-ASSET-02: Format 2 — Downloadable Notes (PDF)

- **Test Steps:** Click "**PDF Summary Notes**" icon in chapter breakdown.
- **Expected Outcome:** Opens downloadable PDF viewer window.

TC-ASSET-03: Format 3 — Interactive MCQ Quiz Engine

- **Test Steps:**

1. Click "**Test MCQs**". 2. Answer 5 questions with options (A, B, C, D). 3. Submit Quiz.

- **Expected Outcome:** Real-time score summary (e.g., 4/5 - 80%), timer breakdown, and instant step-by-step explanations for each question.

TC-ASSET-04: Format 4 — Supplementary PDF Attachments

- **Test Steps:** Open attachments tab → View worksheet download link.
- **Expected Outcome:** PDF lab worksheet accessible for offline revision.

TC-ASSET-05: Format 5 — DRM Video Lectures

- **Test Steps:** Click "**Play Video Class**" button.
- **Expected Outcome:** Video player launches with playback speed controls (1.0x, 1.25x, 1.5x) and DRM stream verification.

Suite 5: Standalone E-Book Store & Instant Delivery

TC-EBOOK-01: Standalone E-Book Catalog Browsing

- **Test Steps:** Navigate to E-Book Store → Verify books: *KAS Prep Handbook (₹50)*, *Indian Constitution (₹100)*.
- **Expected Outcome:** Catalog displays book title, cover, author, rating, page count, and standalone price without requiring full course subscription.

TC-EBOOK-02: E-Book Wallet Purchase & Instant Unlocking

- **Test Steps:**

1. Click **"Buy E-Book for ₹199"** on *Indian Polity Handbook*. 2. Confirm wallet debit.

- **Expected Outcome:** Wallet balance deducted by ₹199; transaction logged; book status instantly changes to **"PURCHASED"** ; **"Read PDF"** button unlocked.

Suite 6: In-App Wallet & UPI Payment Engine

TC-WLT-01: Wallet Balance & Ledger Display

- **Test Steps:** Open Wallet tab → Review Current Balance & Recent Transactions.
- **Expected Outcome:** Displays formatted INR balance and credit/debit list with reference IDs.

TC-WLT-02: Razorpay UPI Wallet Top-Up

- **Test Steps:** Click **"+ Top-Up Wallet"** → Enter amount (**₹2,500**) → Confirm payment.
- **Expected Outcome:** Wallet balance increases by ₹2,500; **CREDIT** transaction created in database.

Suite 7: Binary MLM Network Topology & Placement Engine

TC-MLM-01: Two-Leg Binary Tree Visualization (Depth 4)

- **Test Steps:** Navigate to **"Affiliate Downline Tree"** on Web Portal.
- **Expected Outcome:** Visual hierarchical binary tree renders Root node, Left Leg node (L1), Right Leg node (R1), and Level-2 sub-nodes (L-L, L-R, R-L, R-R) with rank badges and PV metrics.

TC-MLM-02: Downline Leg Placement Preference Selector (**AUTO** / **LEFT** / **RIGHT**)

- **Test Steps:**

1. Select placement preference **"LEFT"**. 2. Register a new user using referral link.

- **Expected Outcome:** New user automatically placed at extreme left vacant spot in binary tree.

TC-MLM-03: Auto-Balance Placement Algorithm

- **Test Steps:** Set placement preference to **"AUTO"** → Register new user.
- **Expected Outcome:** System evaluates left leg volume vs right leg volume and places new user in the weaker volume leg.

Suite 8: Binary Volume Matching, Payout Capping & Deductions Engine

TC-COMM-01: 1:1 Matched Volume & 10% Weaker Leg Payout

- **Pre-conditions:** Left Leg PV = 12,500; Right Leg PV = 9,800.
- **Test Steps:** Trigger payout preview / stats API (`GET /api/mlm/stats`).
- **Expected Outcome:**
 - Matched Volume = `min(12,500, 9,800) = 9,800 PV`
 - Gross Bonus (10%) = `₹980.00`

TC-COMM-02: Carry Forward Volume Calculation

- **Test Steps:** Inspect carry forward values after 9,800 PV matching.
- **Expected Outcome:**
 - Carried Left PV After = `12,500 - 9,800 = 2,700 PV`
 - Carried Right PV After = `9,800 - 9,800 = 0 PV`

TC-COMM-03: Daily Capping Protection Guard (₹25,000)

- **Pre-conditions:** Matched Volume = 300,000 PV (Gross Bonus = ₹30,000).
- **Test Steps:** Run payout calculation engine.
- **Expected Outcome:** Capped Gross Bonus capped at maximum ₹25,000 limit; `isCapped: true`.

TC-COMM-04: Mandatory Deductions (5% Admin + 5% TDS)

- **Pre-conditions:** Capped Gross Bonus = ₹25,000.
- **Test Steps:** Verify net payout breakdown.
- **Expected Outcome:**
 - Admin Fee (5%) = `₹1,250.00`
 - TDS Deduction (5%) = `₹1,250.00`
 - Net Payable Payout = `₹25,000 - ₹2,500 = ₹22,500.00`

Suite 9: Native Mobile Experience & 1-Click Referral Share

TC-MOB-01: Dual Theme System (Light / Dark Mode Toggle)

- **Test Steps:** Open Mobile App → Tap "☀️ Light" / "🌙 Dark" toggle button.
- **Expected Outcome:** App theme seamlessly switches background, text, card borders, and metrics colors without layout glitch.

TC-MOB-02: 1-Click Clipboard Link Copy

- **Test Steps:** Tap "📄 Copy Link" in MLM tab.
- **Expected Outcome:** Copies referral link (`http://:3000/register?ref=EDU-99201`) to device clipboard; displays alert toast.

TC-MOB-03: OS Native Share Sheet Integration

- **Test Steps:** Tap " **Share Link**" on Android device.
- **Expected Outcome:** Opens Android native share sheet allowing 1-click sharing to WhatsApp, Telegram, SMS, and Email.

Suite 10: Admin Panel & Settlement Execution Engine

TC-ADM-01: Batch Binary MLM Payout Settlement Cycle

- **Test Steps:** Log in as Super Admin → Click " **Run Payout Settlement**".
- **Expected Outcome:** Executes settlement for all accounts with matched PV; credits net payouts to user wallets; updates carry-forward volumes; logs transaction and **MlmPayout** audit records.

3. Requirement Traceability Matrix (RTM)

Requirement Description	Specification Ref	Primary Code Module	Verifying UAT Test Case IDs
User Login & Registration	Sec 3.1	<code>authController.js</code> , <code>AuthScreen.tsx</code>	TC-AUTH-01, TC-AUTH-02
State Regional Localization	Sec 3.1	<code>StatePickerModal.tsx</code> , <code>State.js</code>	TC-AUTH-03
Aadhaar & PAN KYC Scanner	Sec 3.1	<code>kycController.js</code> , <code>StudentHomeScreen.tsx</code>	TC-KYC-01, TC-KYC-02, TC-KYC-03
1-Year Subscription Period	Sec 3.1	<code>Purchase.js</code> , <code>purchaseController.js</code>	TC-AUTH-04
6 Core Verticals Taxonomy	Sec 2	<code>ContentHierarchy.tsx</code> , <code>mockData.ts</code>	TC-EDU-01 to TC-EDU-06
5 Primary Asset Formats	Sec 3.3	<code>LearningAsset.js</code> , <code>McqQuizEngine.tsx</code>	TC-ASSET-01 to TC-ASSET-05
Standalone E-Book Store	Sec 3.4	<code>ebookController.js</code> , <code>WalletEbookStore.tsx</code>	TC-EBOOK-01, TC-EBOOK-02
In-App Digital Wallet	Sec 3.5	<code>walletController.js</code> , <code>Wallet.js</code>	TC-WLT-01, TC-WLT-02

Requirement Description	Specification Ref	Primary Code Module	Verifying UAT Test Case IDs
Two-Leg Binary Tree Topology	Sec 4.1	MLMNode.js , MlmTreeVisualizer.tsx	TC-MLM-01, TC-MLM-02, TC-MLM-03
1:1 Pair Match & 10% Bonus	Sec 4.2	mlmController.js , mlmService.js	TC-COMM-01
Carry Forward PV Roll-over	Sec 4.2	mlmController.js , MLMNode.js	TC-COMM-02
₹25,000 Daily Capping Guard	Sec 4.3	mlmController.js , MlmPayout.js	TC-COMM-03
5% Admin + 5% TDS Deductions	Sec 4.3	mlmController.js , AdminPanel.tsx	TC-COMM-04
Mobile Dual Theme & Share	Sec 1.2	StudentHomeScreen.tsx	TC-MOB-01, TC-MOB-02, TC-MOB-03
Admin Settlement Engine	Sec 3.5	mlmController.js	TC-ADM-01

Approval Signature: _____ (Lead QA Architect) **Date of Execution:** 04 / September / 2026